

Vantage MRI PM Report



Document #: TMS-02-00-PM2

Revision: A

Affected Products

Vantage MRI

Reference Material

2Y900-038E*C, Maintenance and Inspection Manual for Superconducting MRI System Toshiba MRI Excelart Series Models MRT-1503/PD, 13, R3, P2, Q2, R2

History

Date	Name	Rev	Change Descriptions
6/9/15	Document Control	A	Initial release

Approval

Department	Name	Date
Dept. Manager		
Engineering		
QA & Safety		
Document Control	Jhovany Hernandez	6/9/15

Preface

This document is valid proof of a Preventative Maintenance performed on the system. The Preventative Maintenance must be performed in the specified intervals, and the results are to be documented herein.

Refer to 2Y900-038E*C, Maintenance and Inspection Manual for Superconducting MRI System Toshiba MRI Excelart Series Models MRT-1503/PD, 13, R3, P2, Q2, R2 for specific procedures.

This section addresses planned maintenance that should be performed by service personnel. Some of it is also listed in the Operator's Manual for the user to perform on daily and weekly basis.

However, service personnel should make sure that these procedures have been followed and if not, include them on the periodic visits.

Warning: All steps must be performed in accordance with the manufacturer's service manual.

This document must be completely filled out by the Field Service Engineer, i.e.:

- All boxes must be filled out. Use N/A where appropriate.
- Enter the Serial Number and the Date in the header of each page.
- After completing the check, fill out and sign the last page.

Equipment Needed

Serial #:

Date:

System Information

Customer:

Model:

System S/N:

System ID:

Measuring Equipment Used

Enter the measuring equipment used in the table below (meters, scopes, phantoms, etc.):

Equipment:

Serial #:

Serial #:

Date:

Checklist

Controller

Pass	Fail	N/A	Action
☐	☐	☐	Clean Air filters at the left and right side covers of the cabinet
☐	☐	☐	Clean inside the cabinet
☐	☐	☐	Clean the fan at the rear of the cabinet
☐	☐	☐	Check the looseness of the terminal
☐	☐	☐	Perform a fan rotation check -Rear of the cabinet -Top the the RM Chassis
☐	☐	☐	Checking the DC power supply
☐	☐	☐	Replacing the UPS
☐	☐	☐	Cleaning the mouse

Gradient Power Supply Section of Control Cabinet

Pass	Fail	N/A	Action
☐	☐	☐	Clean the cabinet
☐	☐	☐	Looseness of the terminal
☐	☐	☐	Check the water filter
☐	☐	☐	Check fan rotation
☐	☐	☐	Check for water leaks
☐	☐	☐	Check cooling water presser
☐	☐	☐	Check the cooling water temperature
☐	☐	☐	Check the water flow rate

Serial #:

Date:

RF Cabinet

Pass	Fail	N/A	Action
☐	☐	☐	Check the AC power Supply input voltage
☐	☐	☐	Check the air intake/outlet of the RF amplifier
☐	☐	☐	Check the fan rotation
☐	☐	☐	Check the RF unit voltage
☐	☐	☐	Check the cooling fans in the RF chassis
☐	☐	☐	Check the cooling fans in the RF cabinet

Oxygen Monitor

Pass	Fail	N/A	Action
☐	☐	☐	Check the oxygen concentration
☐	☐	☐	Check the LOBAT indication
☐	☐	☐	Perform operational checks
☐	☐	☐	Check calibration and sensor life
☐	☐	☐	Perform battery replacement
☐	☐	☐	Perform sensor replacement

Serial #:

Date:

Refrigerator

Pass	Fail	N/A	Action
☐	☐	☐	Check the cooling water for water leaks
☐	☐	☐	Check cooling water pressure (on the IN side)
☐	☐	☐	Check cooling water temperature
☐	☐	☐	Check cooling water flow rate
☐	☐	☐	Check the pressure of the compressor
☐	☐	☐	Replace the displacer for the cold head
☐	☐	☐	Replace the adsorbent for the compressor

Gradient Coil

Pass	Fail	N/A	Action
☐	☐	☐	Check the cooling water pressure P1-P2
☐	☐	☐	Check the cooling water pressure on the IN side of the gradient coil
☐	☐	☐	Check for water leak
☐	☐	☐	Check the water temperature
☐	☐	☐	Check the operation of the flow switch
☐	☐	☐	Check in the flow switch unit

Serial #:

Date:

Gantry

Pass	Fail	N/A	Action
☐	☐	☐	Check the external appearance of the connector port
☐	☐	☐	Check the lamp in the gantry
☐	☐	☐	Check the cooling fan in the gantry
☐	☐	☐	Check the projector
☐	☐	☐	Perform an operation check of the emergency switches
☐	☐	☐	Replace the patient-call tube air-supply bulb
☐	☐	☐	Check for looseness of the connection section of the current terminals of the gantry terminal block

Magnet

Pass	Fail	N/A	Action
☐	☐	☐	Check the amount of evaporation of liquid helium for OR76-type magnet
☐	☐	☐	Check the amount of stored liquid helium
☐	☐	☐	Check the liquid helium container of the OR76-type magnet for leaks

Wall Cabinet

Pass	Fail	N/A	Action
☐	☐	☐	Perform an operational check of the cooling fan
☐	☐	☐	Check the operation of the couch
☐	☐	☐	Checking the operation of the temperature sensor

Serial #:

Date:

Patient Couch

Pass	Fail	N/A	Action
☐	☐	☐	Check the free operation of the couctop
☐	☐	☐	Check the hydraulic hose, clamp section
☐	☐	☐	Check the cable clamp section
☐	☐	☐	Clean the interior of the cover
☐	☐	☐	Clean the couctop travel section
☐	☐	☐	Check the tension of the belt
☐	☐	☐	Check for oil leaks in the hydraulic system
☐	☐	☐	Check the hydraulic cylinder section
☐	☐	☐	Check the connection section of the cylinder and the hose
☐	☐	☐	Check the lubrication oil, up/down movement guide rail, roller, link
☐	☐	☐	Check the looseness of the tightening section
☐	☐	☐	Replacing the accessories of the table
☐	☐	☐	Perform operational check of the couctop limits
☐	☐	☐	perform operational check of the couctop sensor

RF Coils

Pass	Fail	N/A	Action
☐	☐	☐	Perform a QD whole-body coil tuning check
☐	☐	☐	Perform a QD whole-body coil image quality check using phantom
☐	☐	☐	Check the external appearance of all other coils
☐	☐	☐	Check the external appearance of the connector and cable

Serial #:

Date:

Transformer Cabinet

Pass	Fail	N/A	Action
☐	☐	☐	Perform a voltage check of each output section
☐	☐	☐	Perform operational check of the emergency switch
☐	☐	☐	Check the vacuum pump for abnormal sounds
☐	☐	☐	Check the vacuum pump for water
☐	☐	☐	Replace the vacuum pump diaphragm and bearing

Supervisory unit

Pass	Fail	N/A	Action
☐	☐	☐	Check the indicators on the D1 to D4 modules
☐	☐	☐	Replace the battery pack

Entire System

Pass	Fail	N/A	Action
☐	☐	☐	Check the F_0 value
☐	☐	☐	Check the waveform of the gradient magnetic field
☐	☐	☐	Check the SEP waveform
☐	☐	☐	Check the ADM offset
☐	☐	☐	Check the slice thickness
☐	☐	☐	Checking T_2 {AX, SG, CR
☐	☐	☐	Check the S/N of each coil
☐	☐	☐	Check, the sensitivity nonuniformity
☐	☐	☐	Check resolution readout and encoding direction
☐	☐	☐	Check power compensation by additional cabling

Serial #:

Date:

System Condition

After performing scheduled PM, an evaluation of the system is performed with the following result (check one):

- ☐ No deficiencies. The image quality test resulted in no differences from required reference values.
- ☐ Slight deficiencies. There is no effect on operation; however they should be corrected preventively. The image quality test resulted in no differences from required reference values.
- ☐ Serious deficiencies. This must be corrected before operation of the system is permitted.

Field Service Engineer

Name:

Date:

Notes (Additional work to be performed, partial PM, customer requests):

Customer

Name:

Date: